

LAB WEEK 2

DESCRIPTIVE STATISTICS

SMOKING AND DRINKING CESSATION TRIAL

Pharmacological treatments that can concomitantly address cigarette smoking and heavy drinking stand to improve health care delivery for these highly prevalent co-occurring conditions. This superiority trial compared the combination of varenicline and naltrexone against varenicline alone for smoking cessation and drinking reduction among heavy-drinking smokers.



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Ray, L.A., Green, R., Enders, C., et al. (2021). Efficacy of combining varenicline and naltrexone for smoking cessation and drinking reduction: A randomized clinical trial. *American Journal of Psychiatry*, 178, 818–828.

LOAD PACKAGES AND IMPORT DATA

-  = data frame name
-  = variable name
-  = raw data file name

```
# LOAD R PACKAGES ----
```

```
# load R packages
library(ggplot2)
library(psych)
library(summarytools)
```

```
# READ DATA ----
```

```
# github url for raw data
filepath <-
  'https://raw.githubusercontent.com/craigenders/psych250a/main/data/ClinicalTrialData.csv'
```

```
# create data frame called ClinicalTrial from github data
ClinicalTrial <- read.csv(filepath, stringsAsFactors = T)
```


SUMMARIZING DATA

 = data frame name
 = variable name

```
# INSPECT DATA ----
```

```
# summarize entire data frame (summarytools package)  
dfSummary()
```

```
# DESCRIPTIVE STATISTICS ----
```

```
# descriptive statistics for entire data frame (psych package)  
describe()
```

R OUTPUT

Data Frame Summary
ClinicalTrial
Dimensions: 165 x 2
Duplicates: 125

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	Condition [factor]	1. Varenicline	82 (49.7%)	IIIIIIIIIII	165	0
		2. Varenicline + Naltrexone	83 (50.3%)	IIIIIIIIIIII	(100.0%)	(0.0%)
2	COWeek8 [integer]	Mean (sd) : 5.5 (6)	25 distinct values	:	165	0
		min < med < max:		:	(100.0%)	(0.0%)
		0 < 3 < 29		:		
		IQR (CV) : 8 (1.1)		: .		
				: : :		

R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Gender*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37

SUMMARIES BY GROUP

-  = data frame name
-  = variable name
-  = grouping variable

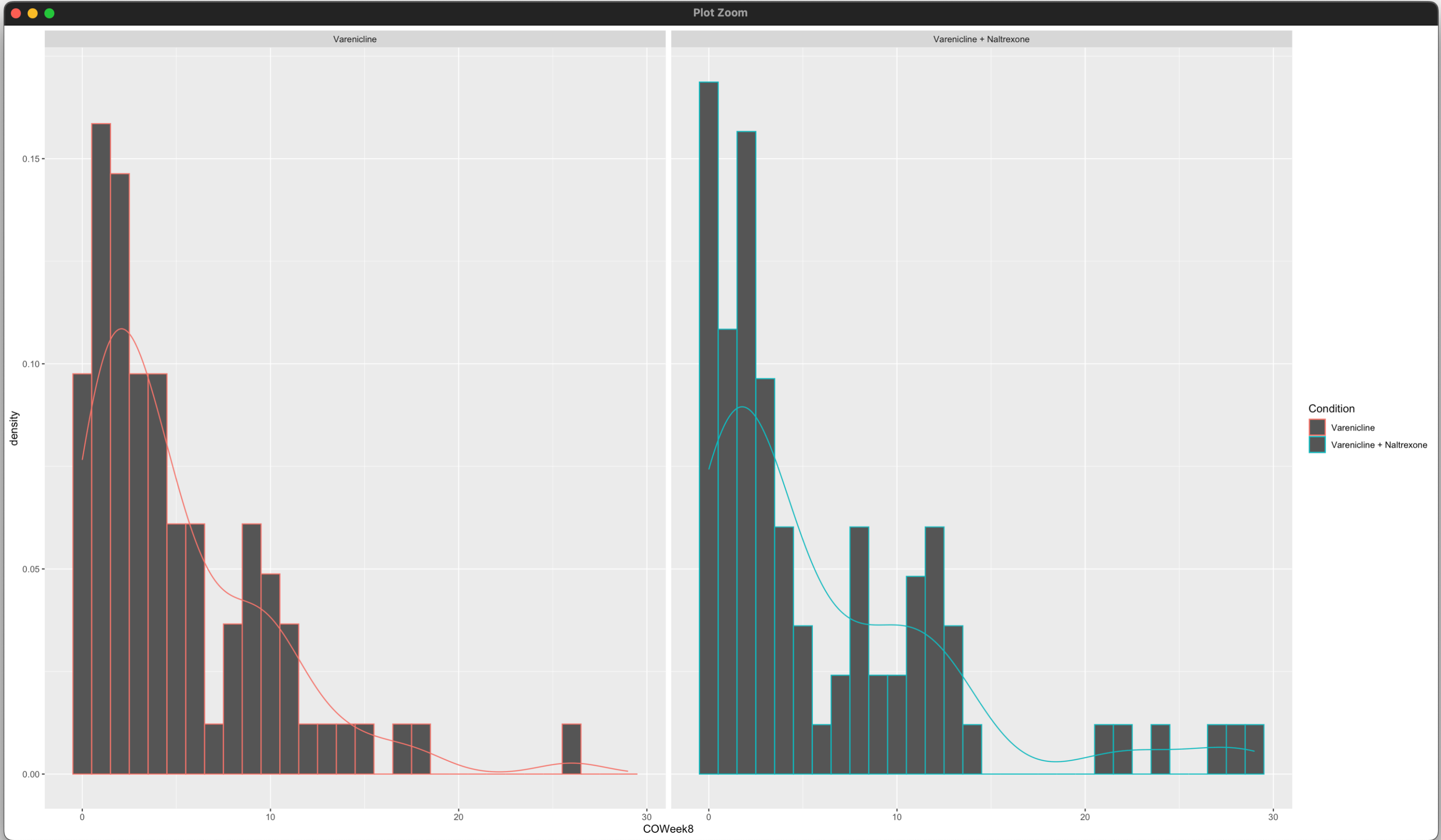
```
# HISTOGRAMS AND KERNEL DENSITY PLOTS BY GROUP ----
```

```
# histogram and kernel density plot separately by group (ggplot2 package)
ggplot(ClinicalTrial, aes(x = COWeek8, colour = Condition)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30) +
  geom_density() +
  facet_wrap(~ Condition)
```

```
# DESCRIPTIVE STATISTICS BY GROUP ----
```

```
# descriptive statistics separately by group (psych package)
describeBy(COWeek8 ~ Condition, data = ClinicalTrial)
```

DEPRESSION DISTRIBUTIONS



R OUTPUT

Descriptive statistics by group
Condition: Varenicline

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
COWeek8	1	82	5.02	4.88	3.5	4.33	3.71	0	26	26	1.57	3.09	0.54

Condition: Varenicline + Naltrexone

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
COWeek8	1	83	6.02	6.86	3	4.81	4.45	0	29	29	1.62	2.31	0.75

STANDARDIZING USING z-SCORES

-  = data frame name
-  = variable name
-  = grouping variable

```
# STANDARDIZING VARIABLES AS Z-SCORES ----
```

```
# convert to z-scores (base R)
```

```
ClinicalTrial$zCOWeek8 <- scale(ClinicalTrial$COWeek8)
```

```
# descriptive statistics for a subset of variables (psych package)
```

```
describe(ClinicalTrial[,c('COWeek8', 'zCOWeek8')])
```

```
# descriptive statistics separately by group (psych package)
```

```
describeBy(zCOWeek8 ~ Condition, data = ClinicalTrial)
```

R OUTPUT

```
> describe(ClinicalTrial[,c('COWeek8','zCOWeek8')])
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
COWeek8	1	165	5.53	5.96	3.00	4.55	2.97	0.00	29.00	29.00	1.75	3.35	0.46
zCOWeek8	2	165	0.00	1.00	-0.42	-0.16	0.50	-0.93	3.94	4.87	1.75	3.35	0.08

```
> describeBy(zCOWeek8 ~ Condition, data = ClinicalTrial)
```

Descriptive statistics by group
Condition: Varenicline

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
zCOWeek8	1	82	-0.08	0.82	-0.34	-0.2	0.62	-0.93	3.44	4.36	1.57	3.09	0.09

Condition: Varenicline + Naltrexone

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
zCOWeek8	1	83	0.08	1.15	-0.42	-0.12	0.75	-0.93	3.94	4.87	1.62	2.31	0.13



SMALL GROUP EXERCISE

Download two files from Bruin Learn: “Week 2 Lab. Descriptive Statistics.R” and “Week 2 Small Group Exercise.R”. The Lab script contains the R code we just discussed. The Exercise script contains only the URL for a different data set, BodySatData.csv. In groups of two or three, you will complete a series of R tasks that provide practice for the next assignment. There is no need to write code from scratch; instead, you can copy and paste code chunks from the Lab file into your Exercise script, modifying the data and variable names as needed. The BodySatData.csv file for this exercise contains body satisfaction data from two age cohorts.

SKIN COLOR SATISFACTION AND BINGE EATING

Although it has been demonstrated that (a) body dissatisfaction and internalization of societal appearance standards contribute to disordered eating and (b) that internalization of societal appearance standards leads to decreased skin color satisfaction among Black women, it has not been established whether skin color dissatisfaction contributes to disordered eating among Black women or girls. The objective of the present study is to determine the influence of skin color satisfaction as a potential predictor for binge eating, and its effect through body image in Black girls during the vulnerable developmental period of adolescence.

Parker, J.E., Enders, C.K., Mujahid, M.S., Laraia, B.A., Epel, E.S., Tomiyama, A.J. (2022). Prospective relationships between skin color satisfaction, body satisfaction, and binge eating in Black girls. *Body Image, 41*, 342-353.



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SMALL GROUP EXERCISE TASK 1

- Use the provided URL to import the BodySatData.csv file into an R data frame (import method #3 from the Week 0 lab script).
- Use the dfSummary function to get numeric and visual summaries of the data frame's variables.
- Use the describe function to get descriptive statistics for the entire sample.



SMALL GROUP EXERCISE TASK 2

- Consider the descriptive statistics for BodySat variable. Why do the two measures of central tendency (median and mean) differ? Your explanation should integrate properties of the distribution (e.g., skewness, presence of outliers) and their effect on each measure.
- Interpret the BodySat standard deviation in practical terms. Your explanation should use the numeric value from your analysis, and it should be understandable to a non-technical audience without losing accuracy.



SMALL GROUP EXERCISE TASK 3

- Use the `describeBy` function from the `psych` package to obtain descriptive statistics for the `BodySat` variable within each of the two age cohorts (Age; 10 versus 18 year olds).
- Evaluate the size of the age mean difference on the `BodySat` variable. How would you gauge the magnitude of the difference? Do girls at ages 10 and 18 exhibit a meaningful difference? Why or why not?